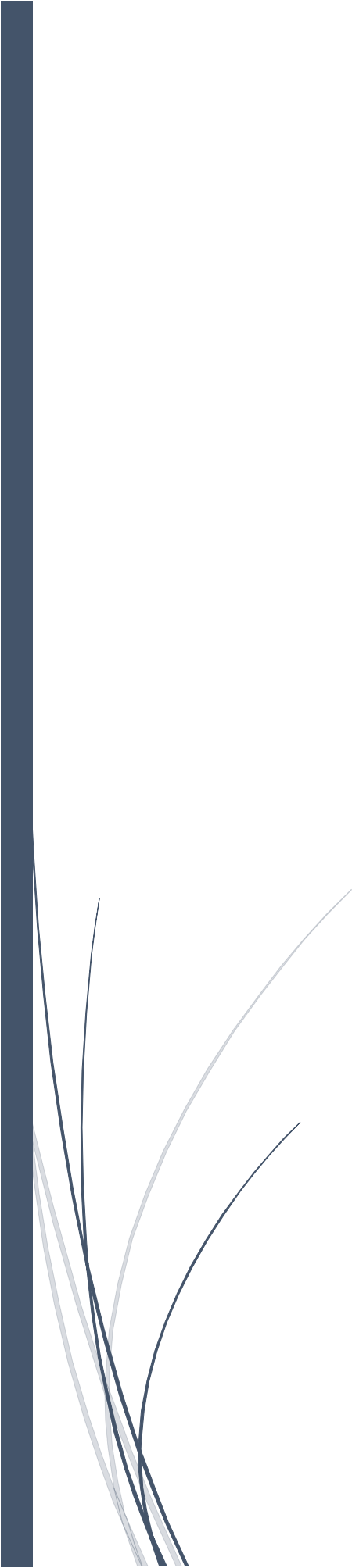


A thick dark blue vertical bar is positioned on the left side of the page. From its base, several thin, curved lines in dark blue and light grey extend upwards and outwards, creating an abstract, organic shape.

EMI Filter Design Theoretical And Practical Approach

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ELARABY GROUP

Contents:

- Introduction
- Importance of EMI Filter
- Noise source and types
- Designing steps of EMI Filter (theoretical and practical)
- practical example on an existing product

Introduction

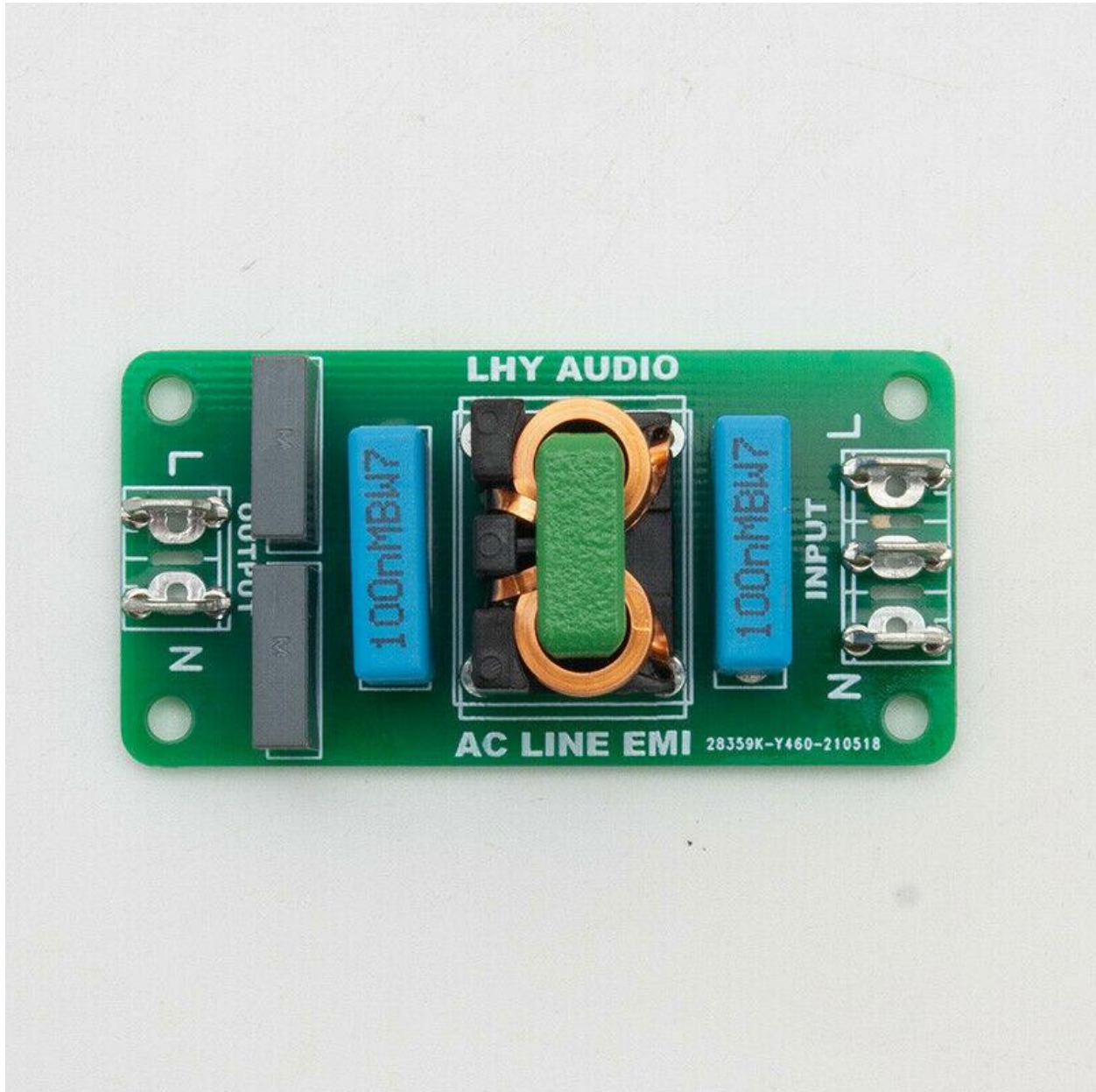


FIGURE 1

Conducted emission regulations are intended to control the radiation from the public alternating current (ac) power distribution system, which results from noise currents conducted back onto the power line. Normally, these currents are too small to cause interference directly with other products connected to the same power line; however, they are large enough to cause the power line to radiate and possibly become a source of interference.

The conducted emission requirements, therefore, are really radiated emission requirements in disguise.

In addition, some products are nonlinear loads on the ac power distribution system. They have nonsinusoidal input current waveforms, rich in harmonics and can have a detrimental effect on the performance of the power distribution system. For this reason, the European Union has regulations limiting the harmonic emission from electronic products

Because it is the product's power supply that directly interfaces to the ac power line, the design of the power supply and the power-line filter has a large influence on the conducted and harmonic emissions. This is especially true in the case of switched-mode power supplies and variable speed motor drives.

Importance of EMI Filter:

In order to make your product vailed to be exported you have to pass an EMC tests the compiled with Europe standards

EMC emissions tests divided into two categories

1. Conducted Emissions
2. Radiated Emissions

There is another category the is another branch of EMC test which is immunity

For exporting you have to pass all these tests.

Notice: that every country has its own standards but it is almost taken from European Union standards which mean if you pass these standards you can export to any country that uses the same standards.

Other countries like Egypt does not have any standards about EMC for that reason you may notice that the products that enters Egypt does not necessarily have EMI filter.

The EMI Filter is mainly designed for a conducted Emission test of the EMC tests which is the disturbance voltage on both L and N.

In this article we will talk only about EMI design and it influences if you need to know more about the EMC I highly recommend to read the book attached in references section.

Noise source and types:

Today's highly sophisticated switched-mode power supplies (SMPS) operate at efficiencies of 85% or greater and are only a fraction of the size and weight of comparable linear supplies. However, all these advantages come with many disadvantages. For one, switched-mode power supplies are a major source of both conducted and radiated emissions. They conduct large noise currents (both common- and differential-mode) back onto the power line at harmonics of the switching frequency

This SMPS configuration has multiple sources of noise, however. Some of these are the result of the normal operation of the circuit (differential-mode noise), whereas others are the result of circuit's parasitic capacitances (common-mode noise)

Common-mode noise:

The major contributor to common-mode emission is the primary side parasitic capacitance to ground. The three contributors to this capacitance are

1. switching transistor to heat sink capacitance
2. transformer interwinding capacitance
3. stray primary side wiring capacitance

as shown below

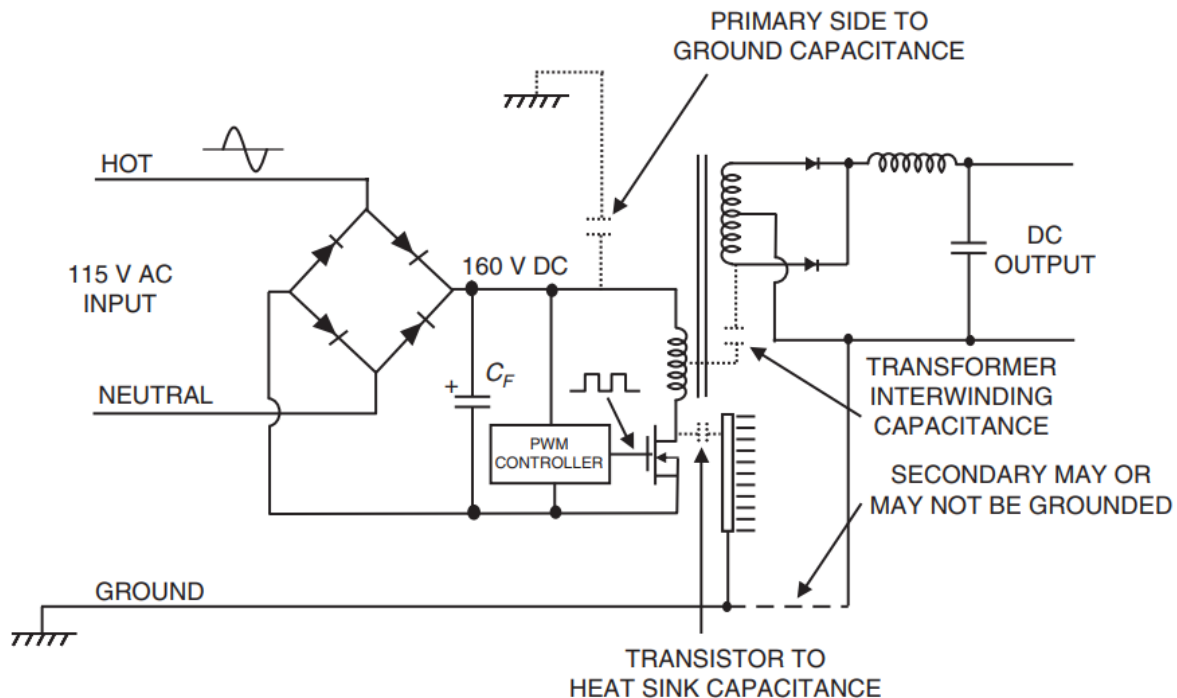


FIGURE 2

Differential-mode noise:

The normal operation of the power supply consists of the switching transistor driving a current at the switching frequency around the loop that consists of the switching transistor, the transformer, and the filter capacitor C_F . As long as the switching current flows through this loop, internal to the power supply, it will produce no differential-mode emission. The primary purpose of the capacitor C_F , however, is to filter the full-wave rectified ac line voltage. The filter capacitor is therefore a large-value, high voltage capacitor (typically 250 to 1000 mF with a voltage rating of 250 V or more), and it is far from an ideal capacitor. It typically has a significant equivalent series inductance (ESL) L_F and equivalent series resistance (ESR) R_F . As a result of these parasitic impedances, not all the switching current will flow through the capacitor C_F . Rather, a current division will occur at the capacitor terminals, with some of the switching current flowing through the capacitor and the remainder flowing through the full-wave bridge rectifier out onto the power line as shown below

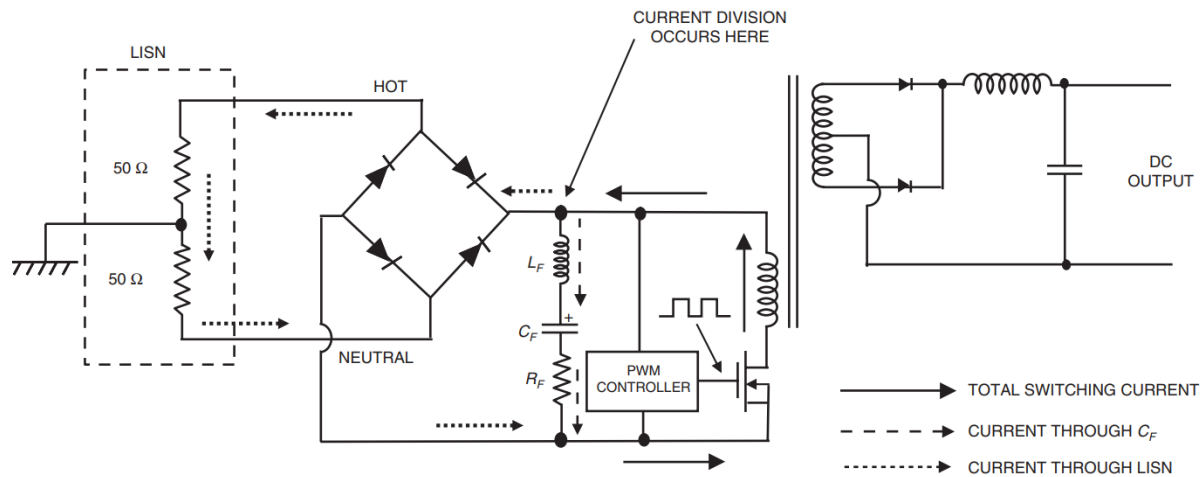


FIGURE 3

Designing steps of EMI Filter:

1. Which noise source is the dominant (common mode or differential mode)

In order to know which type of the noise is the dominant problem you have to determine the noise voltage first

In case of common mode voltage, you can determine the value of noise in voltage with the law below

$$\text{Eq (1) } V_{CM} = 100 \cdot V_P \cdot F_0 \cdot C_P$$

Where V_P is the voltage of the bulk capacitor which is after the rectifier

F_0 is the switching frequency of the power IC

C_P is the parasitic capacitance in the circuit which unfortunately we have to assume it

The common-mode current, and therefore the LISN voltage, are determined primarily by the magnitude of this parasitic capacitance. Typical values for C_P range from about 50 pF to as much as 500 pF.

FIGURE 4

An example is below

Example 1-1. For the case where $V_P = 160 \text{ V}$, $C_P = 200 \text{ pf}$, $t_r = 50 \text{ ns}$, and $F_0 = 50 \text{ kHz}$, the envelope of the common-mode conducted emission will have a magnitude of **160Mv**

(160 milli volt)

In case of differential mode noise voltage, you can determine the value of noise in voltage with the law below

$$\text{Eq (2) } V_{DM} = 2 \cdot F_0 \cdot L_F \cdot I_P$$

Where

F_0 is the switching frequency of the power IC

L_F is the parasitic inductance ESL of the bulk capacitor (from data sheet)

I_P is the maximum current that will be drawn of the power supply

Example 2-1

$I_P = 4\text{ A}$, $L_F = 30\text{ nH}$, $t_r = 50\text{ ns}$, and $F_0 = 50\text{ kHz}$. Neglecting the ESR of the input ripple filter capacitor, the envelope of the differential mode conducted emission will have a magnitude of **12 mV**

Determine the gain of the two results $160\text{ mV}/12\text{ mV} = 13.33$ from this web site link

The dB is equal to 22.4 as shown below from this web site [link](#)

The screenshot shows a web browser window with the URL allaboutcircuits.com/tools/decibel-calculator/. The page title is "Decibel Calculator". It features a navigation bar with links like ARTICLES, FORUMS, EDUCATION, TOOLS, VIDEOS, DATASHEETS, GIVEAWAYS, PARTNER CONTENT HUB, PODCAST, LATEST, and SUBSCRIBE. The main content area has a heading "Decibel Calculator" and a description: "This calculator will convert voltage or power gains into decibels. Enter one of the three values and the calculator will output the other two." Below this, there are three input fields: "Voltage Gain (Vout/Vin):" with the value 13.333333, "Power Gain (Pout/Pin):" with the value 177.777777, and "Decibels (dB):" with the value 22.498775. There are "Calculate" and "Reset" buttons. To the right, there is a "Related Tools" section with links to various calculators and a "FEATURED RESOURCE" section with a link to a white paper titled "Support for SMT Line Setup". The bottom of the page shows a Windows taskbar with various application icons and a system tray displaying the temperature as 41°C and the date/time as 4:43 PM on 6/14/2024.

Comparing the results of Example 2-1 to those of Example 1-1, we observe that in this case the common-mode noise voltage is more than **22.4 dB** greater than the differential-mode noise voltage, **and it is the predominate emission**

Equation 1 shows that the common-mode conducted emission is directly proportional to the magnitude of the switching voltage, whereas Eq. 2 shows that the differential-mode conducted emission is directly proportional to the magnitude of the switching current. Therefore, **in high voltage low-current supplies, the common mode emission will predominate, but in low-voltage high-current supplies the differential mode emission will predominate.**

2. Define the required attenuation of the filter

After knowing the dominant that you have to deal with which is almost common mode noise we have to know how far is the noise voltage the limit of the standard

Let's continue with ex 1-1 with the calculated VCM of 164 mV

To find the limit when need to classify the classes first

We are talking about commercial industrial (Home appliances)

From the figure 5 below, we are now talking **Class B**

And the worse limit we need to design our filter below it is **46B μ V**

EN55022 limit lines: conducted emissions

Class A and Class B limits, quasi-peak & average, 15 0kHz–30 MHz

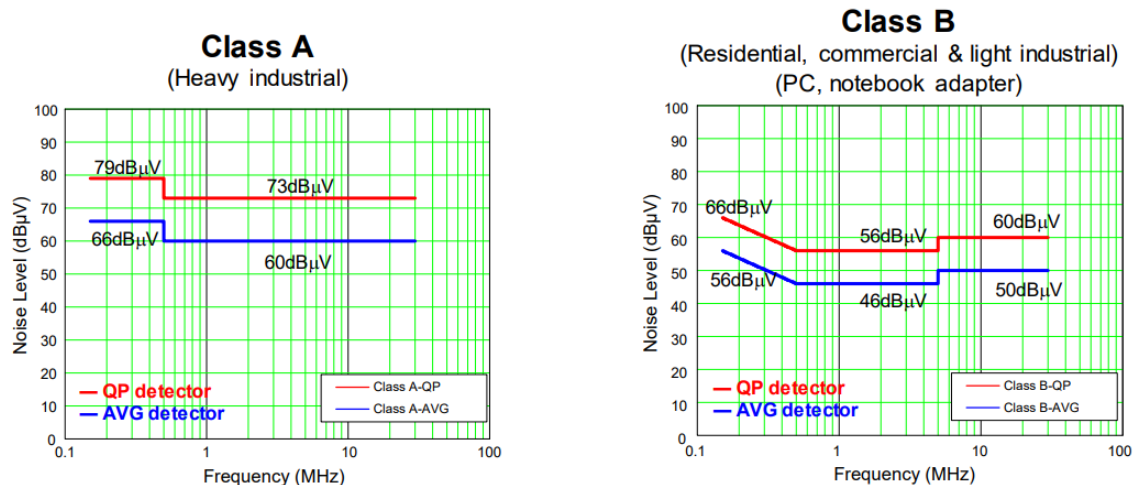


FIGURE 5

we know our dominant noise voltage which is **164mV** from ex 1-1

And the worst limit we need to below it which is 46B μ V

In order to know the required attenuation, we need to convert the limit from dB to V

We can do that with this [website](#)

46B μ V is equal to 199.5 μ V as shown below

EMI FILTER DESIGN THEORETICAL AND PRACTICAL APPROACH

The screenshot shows a web browser with the URL cantwellengineering.com/calculator/convert/dbuv. The page is titled "Convert from dBuV" and includes a search bar. The main content area shows a conversion table for dBuV to various units, with a "Calculate" button. The table lists the following conversions:

Convert from	dBuV with Z =	Calculate
$dB\mu V - 60$	-14.0	$dBmV$
$dB\mu V - 120$	-74.0	dBV
$dB\mu V - 20 \cdot \log_{10}(Z)$	12.0	$dB\mu A$
$dB\mu V - 20 \cdot \log_{10}(Z) - 60$	-48.0	$dBmA$
$dB\mu V - 20 \cdot \log_{10}(Z) - 120$	-108.0	dB
$dB\mu V - 10 \cdot \log_{10}(Z)$	29.0	$dBpW$
$dB\mu V - 10 \cdot \log_{10}(Z) - 90$	-61.0	dBm
$dB\mu V - 10 \cdot \log_{10}(Z) - 120$	-91.0	dBW

Below the table, there are several input fields and their corresponding values:

- $10^{(dB\mu V/20)}$ = 199.5 μV
- $10^{(dB\mu V-60)/20}$ = $199.5 \text{ e-}3$ mV
- $10^{(dB\mu V-120)/20}$ = $199.5 \text{ e-}6$ V
- $10^{(dB\mu V-20 \cdot \log_{10}(Z))/20}$ = 4.0 μA
- $10^{(dB\mu V-20 \cdot \log_{10}(Z)-60)/20}$ = $4.0 \text{ e-}3$ mA
- $10^{(dB\mu V-20 \cdot \log_{10}(Z)-120)/20}$ = $4.0 \text{ e-}6$ A
- $(10^{(dB\mu V/20)})^2$ = 796.2 pW
- $(10^{(dB\mu V/20)})^2 \cdot 10^{-9}$ = $796.2 \text{ e-}9$ mW

On the right side, there is a "RF Converter" section with a list of conversion options and a "Recent Blogs" section with links to "FCC Marketing Rules", "Labeling for EMC", and "CE Marking".

Dividing the common mode noise voltage on worst limit voltage = $164 \text{ mV} / 199.5 \mu V$

Is equal to 822

Convert it to dB as shown below

The screenshot shows a web browser with the URL allaboutcircuits.com/tools/decibel-calculator/. The page is titled "Decibel Calculator" and includes a search bar. The main content area contains a description of the calculator and three input fields for "Voltage Gain (Vout/Vin)", "Power Gain (Pout/Pin)", and "Decibels (dB)". The "Voltage Gain" field is set to 822, and the "Decibels" field is set to 58.297436. There are "Calculate" and "Reset" buttons. Below the calculator, there is a note about the application of the calculator.

Related Tools:

- Instrumentation Amplifier Gain Calculator
- Lorentz Force Calculator
- Parallel Resistor Calculator
- Wheatstone Bridge Calculator
- Electrical Energy Calculator

FEATURED RESOURCE

WHITE PAPERS

Support for SMT Line Setup

Discover how IDENTICO went beyond label support to guide a new customer through the complexities of PC board production, becoming a trusted partner along the way.

Download PDF

The resultant 58 dB is the required attenuation from the EMI Filter you design in order to pass the Dv EMC test

Now we have figured out the required attenuation for EMI Filter design so we have to convert this value into actual filter

Unfortunately, the book role ends here and it is not mentioned how to select the component value of the inductor and capacitors so I have to figure out a way solve this issue

And I have found a way to calculate these values and will be discussed in the practical example

practical example on an existing product:

Now we will implement the theoretical concept into actual EMI Filter board with right chosen of L and C values with live example

We have a product which is a coffee maker that is failing in EMC test which is Dv test

So, I have to design an EMI Filter in order to make the product exported

Let get started.

we know that from explaining above that the dominant noise is the common mode noise and that is our starting point

- 1 Calculate the VCM

$$\text{VCM} = 100 \cdot \text{VP} \cdot \text{F0} \cdot \text{CP}$$

$$\text{VP} = 220 \cdot 1.414 = 310\text{V}$$

$\text{F0} = 68\text{KHZ}$ (from power ic datasheet LNK3206PN in our case)

$\text{CP} =$ I have assumed that value which is 200 pf

$$\text{VCM} = 100 \cdot 310 \cdot 68 \cdot 10^3 \cdot 200 \cdot 10^{-12}$$

$$= 421\text{mv}$$

- 2 Calculate the required attenuation

We know our worst limit which is 46BμV is equal to 199.5μV from the example explained above

So, dividing the VCM on the limit voltage $= 421\text{mv} / 199.5\mu\text{V} = 2110$

Go to the [link](#) to convert the calculated value to dB

The calculated value in dB is 66 dB

The required attenuation from our EMI Filter design is 66 dB

I have searched a lot on the internet to find a document that uses that value to calculate L and C

And found that [document](#)

Ok let get started with second part of the EMI filter design

Our filter will be LC filter which is Second Order Filters lets go to the example of that type in the document above

The steps of choosing the LC values as follow

1) Identify the required cutoff frequency:

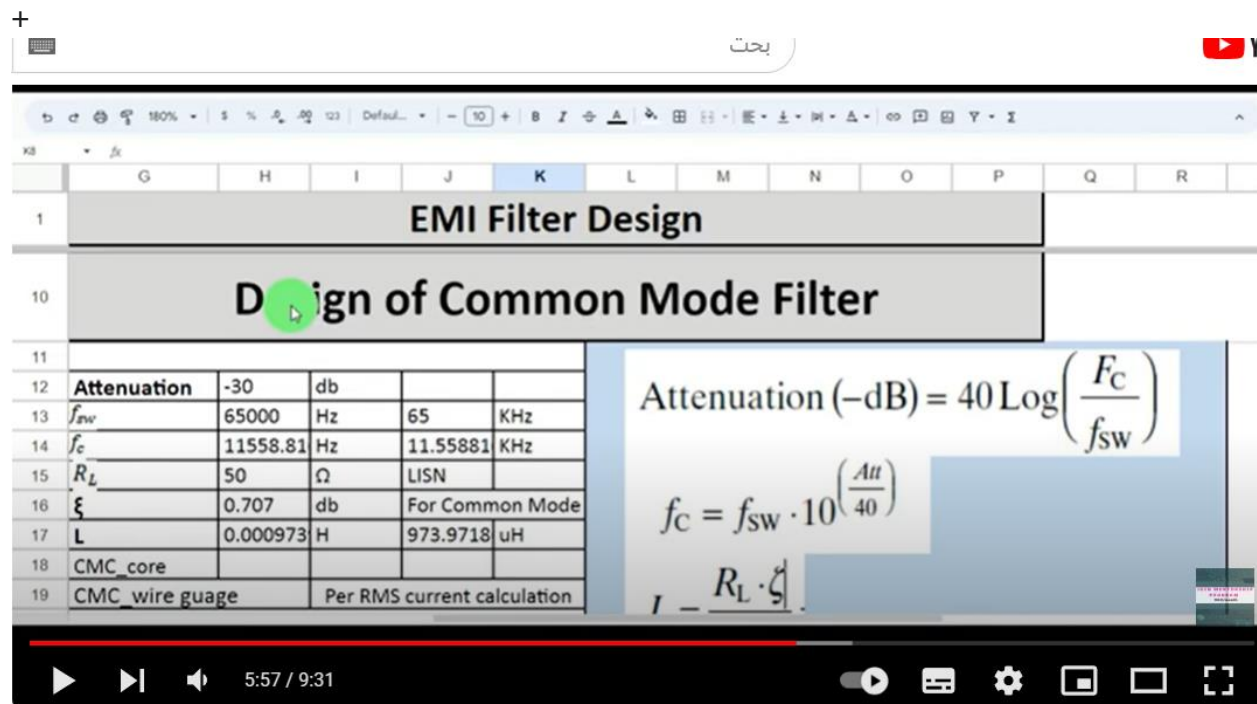
For this example, suppose we have a switching power supply (for use in equipment covered by UL478) that is actually 24 dB noisier at 60 KHz than permissible for the intended application. For a second order filter (12 dB/octave roll off) the desired corner frequency would be 15 KHz.

FIGURE 6

Identify the required cutoff frequency:

Unfortunately, the document doesn't explain how he gets this value so I searched online until I found this law

+



The screenshot shows a YouTube video player displaying an Excel spreadsheet titled "EMI Filter Design". The spreadsheet contains a table with the following data:

	G	H	I	J	K	L	M	N	O	P	Q	R
1	EMI Filter Design											
10	Design of Common Mode Filter											
11												
12	Attenuation	-30	db									
13	f_{sw}	65000	Hz	65	KHz							
14	f_c	11558.81	Hz	11.55881	KHz							
15	R_L	50	Ω	LISN								
16	ξ	0.707	db	For Common Mode								
17	L	0.000973	H	973.9718	μH							
18	CMC_core											
19	CMC_wire guage			Per RMS current calculation								

Next to the table, the following formula is displayed:

$$\text{Attenuation (-dB)} = 40 \log \left(\frac{F_c}{f_{sw}} \right)$$

$$f_c = f_{sw} \cdot 10^{\left(\frac{Att}{40} \right)}$$

The video player interface shows the video is at 5:57 / 9:31.

(I have applied this law in order to get the f_c or corner frequency and applied with the variables that shown in figure 6 in the document to prove if it's true or not and the result is that law is correct and gave me the result of corner frequency which is 15khz as shown in figure 6)

Ok let's get F_C or corner frequency for our case:

$$-66\text{dB} = 40\log(F_C/68\text{KHZ})$$

FC=1522 HZ

2) Identify the load resistance at the cutoff frequency:

Assume $R_L = 50 \Omega$

Identify the load resistance at the cutoff frequency: Assume $R_L = 50 \Omega$ (LISN Impedance)

3) Choose the desired damping factor:

Choose a minimum of 0.707 which will provide 3 dB attenuation at the corner frequency while providing favorable control over filter ringing.

Choose the desired damping factor:

Damping factors much greater than 1 may cause unacceptably high attenuation of lower frequencies whereas a damping factor much less than 0.707 may cause undesired ringing and the filter may itself produce noise

4) Calculate required component values:

$$\omega_n = 2\pi f_n = 94248 \text{ rad / sec}$$

$$C = \frac{1}{L\omega_n^2}$$

$$\zeta = 0.707 = \frac{L\omega_n}{2R_L}$$

Calculate required component values:

$$\omega_n = 2\pi f_n$$

$$\omega_n = 2\pi \cdot 1522 = 9563$$

Calculating L:

$$\zeta = 0.707 = \frac{L\omega_n}{2R_L}$$

$$0.707 = 9563 \cdot L / 2 \cdot 50$$

$$L = 739 \mu\text{H}$$

Calculating Y C:

$$C = \frac{1}{L\omega_n^2}$$

$$= 1 / 739 \mu\text{H} \cdot 9563^2$$

$$= 14.7 \text{ uf}$$

This value is not acceptable because the maximum value of the line-to-ground capacitors is limited because of leakage requirements imposed by various safety agencies. Excessive leakage current to ground is considered a shock hazard and therefore is regulated. Worldwide leakage requirements vary from 0.5 mA to 5 mA according to the application and the safety agency

So, I have searched in Elaraby database for the available Y capacitors and Common choke coils and insert it in the Laws above until I reached to suitable inductance of common mode choke coil and capacitance of Y capacitor

The Y I found is **1nf** whose part number is **MDE1E3KX102MN4**

Starting with this value lets calculate the inductance value of common choke coil

$$C = \frac{1}{L\omega_n^2}$$

$$1\text{nf} = 1 / L \cdot 9563^2$$

$$L = 10.6\text{mH}$$

I have searched for that value and found common choke coli with inductance equal to **10.5μH** which is very close to our calculation whose part number is **RRFiLNA027CBZZ**

Now let's Calculate actual frequency:

$$\frac{1}{2\pi\sqrt{LC}} = 1553.19\text{HZ (very close to the calculated corner frequency above 1522)}$$

The final circuit

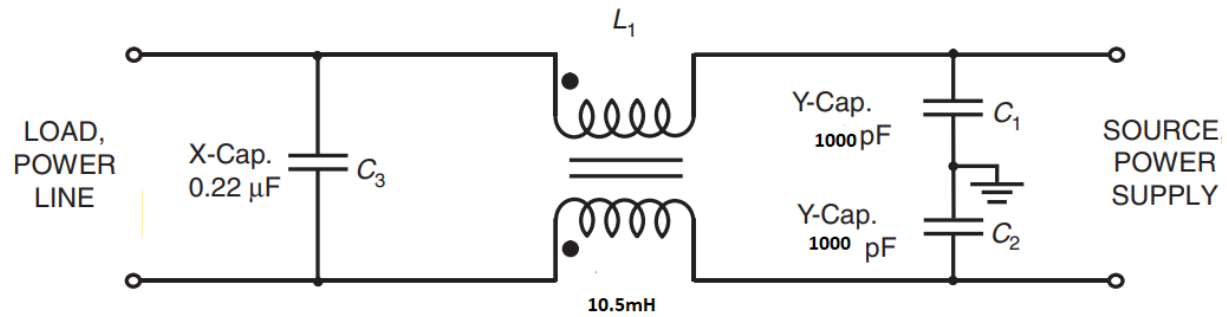


FIGURE 7

NOW LET'S SEE THE RESULT

BEFORE:

• **< Disturbance Voltage of AC Mains port L and N Terminal >**

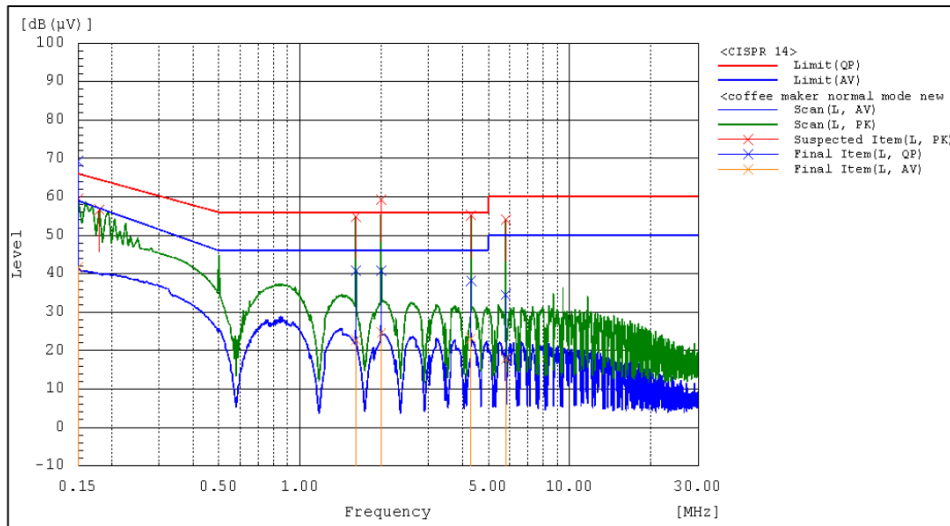


Fig 1: Disturbance Voltage spectrum of L-Terminal .

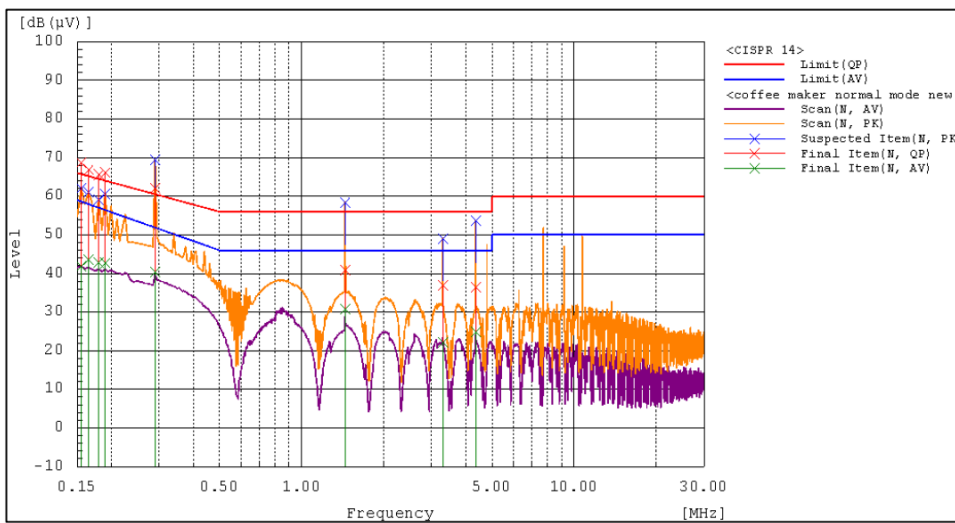


Fig 2: Disturbance Voltage spectrum of N-Terminal .

Final Items Data

Frequency [MHz]	Line	Reading QP [dB(μV)]	Reading AV [dB(μV)]	Factor [dB]	Level QP [dB(μV)]	Level AV [dB(μV)]	Limit QP [dB(μV)]	Limit AV [dB(μV)]	Margin QP [dB]	Margin AV [dB]
0.150	L	59.0	31.6	10.2	69.2	41.8	66.0	59.0	-3.2	17.2
1.610	L	30.5	12.5	10.3	40.8	22.8	56.0	46.0	15.2	23.2
2.000	L	30.6	14.4	10.3	40.9	24.7	56.0	46.0	15.1	21.3
4.305	L	27.9	12.9	10.3	38.2	23.2	56.0	46.0	17.8	22.8
5.800	L	24.2	7.6	10.3	34.5	17.9	60.0	50.0	25.5	32.1
0.155	N	58.6	31.7	10.2	68.8	41.9	65.7	58.6	-3.1	16.7
0.165	N	56.7	33.3	10.2	66.9	43.5	65.2	58.0	-1.7	14.5
0.180	N	55.4	32.7	10.2	65.6	42.9	64.5	57.0	-1.1	14.1
0.190	N	55.9	32.5	10.2	66.1	42.7	64.0	56.4	-2.1	13.7
0.290	N	51.8	30.3	10.2	62.0	40.5	60.5	51.9	-1.5	11.4
1.445	N	30.6	20.6	10.3	40.9	30.9	56.0	46.0	15.1	15.1
3.305	N	26.6	11.8	10.3	36.9	22.1	56.0	46.0	19.1	23.9
4.360	N	26.1	14.6	10.3	36.4	24.9	56.0	46.0	19.6	21.1

AFTER THE EMI:

- **< Disturbance Voltage of AC Mains port L and N Terminal >**

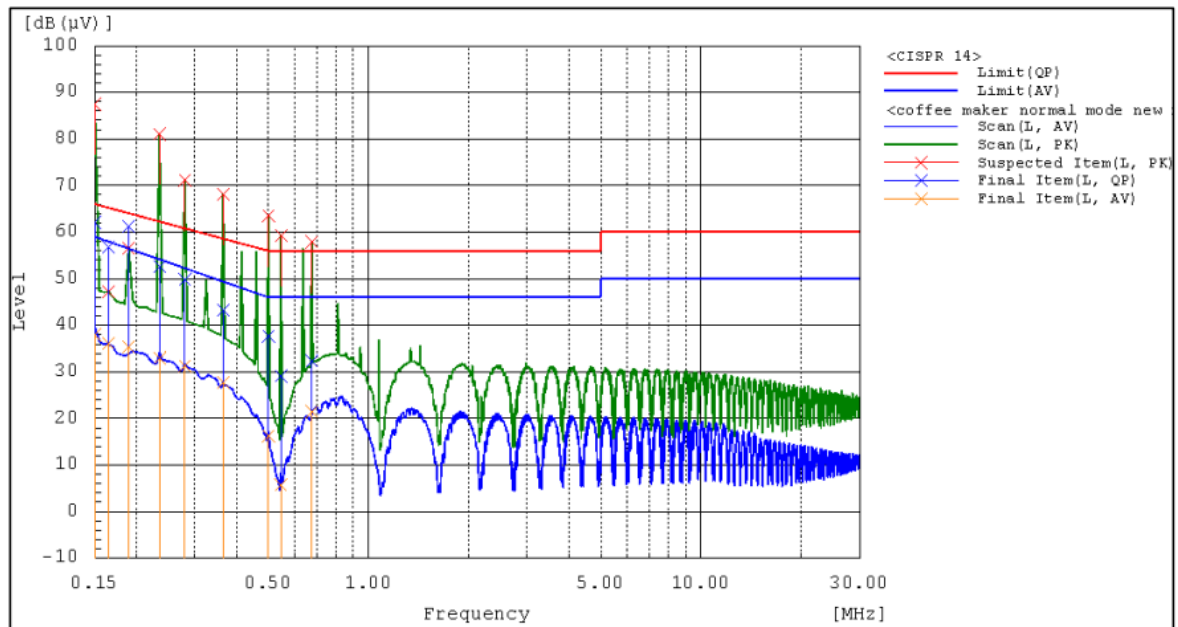


Fig 1: Disturbance Voltage spectrum of L-Terminal.

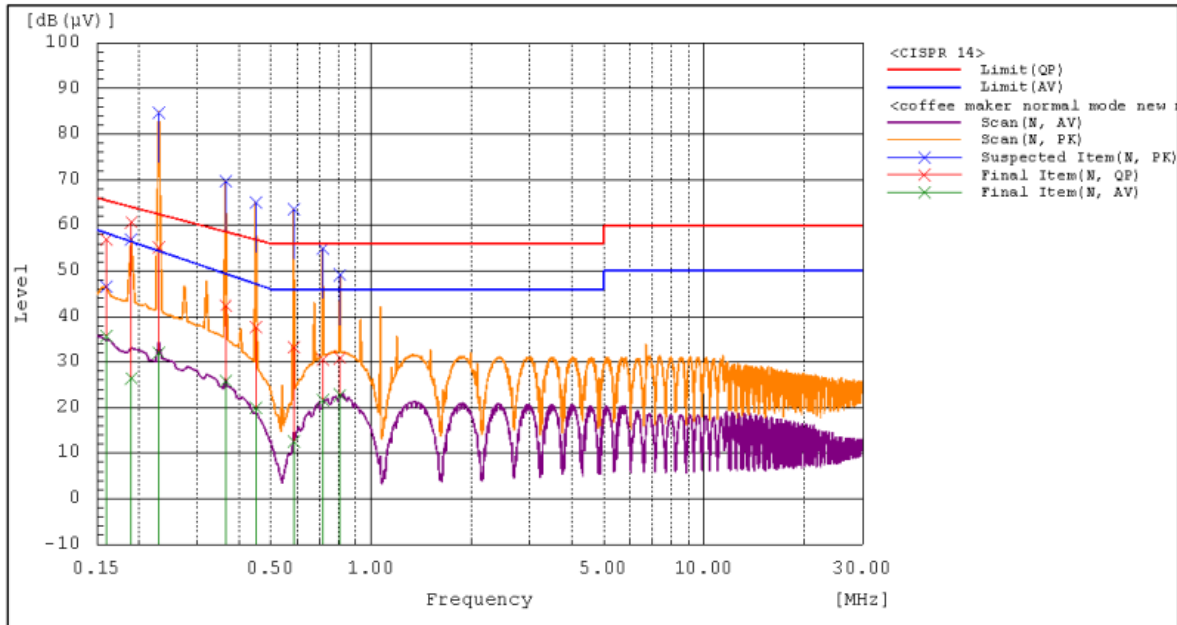


Fig 2: Disturbance Voltage spectrum of N-Terminal.

Final Items Data

Frequency [MHz]	Line	Reading QP [dB(μV)]	Reading AV [dB(μV)]	Factor [dB]	Level QP [dB(μV)]	Level AV [dB(μV)]	Limit QP [dB(μV)]	Limit AV [dB(μV)]	Margin QP [dB]	Margin AV [dB]
0.150	L	51.9	27.8	10.2	62.1	38.0	66.0	59.0	3.9	21.0
0.165	L	46.7	26.1	10.2	56.9	36.3	65.2	58.0	8.3	21.7
0.190	L	51.0	25.2	10.2	61.2	35.4	64.0	56.4	2.8	21.0
0.235	L	42.4	22.8	10.2	52.6	33.0	62.3	54.2	9.7	21.2
0.280	L	39.6	21.1	10.2	49.8	31.3	60.8	52.3	11.0	21.0
0.365	L	33.2	17.6	10.2	43.4	27.8	58.6	49.4	15.2	21.6
0.500	L	27.5	6.1	10.2	37.7	16.3	56.0	46.0	18.3	29.7
0.545	L	18.9	-4.5	10.2	29.1	5.7	56.0	46.0	26.9	40.3
0.675	L	22.2	11.2	10.2	32.4	21.4	56.0	46.0	23.6	24.6
0.160	N	46.7	25.6	10.2	56.9	35.8	65.5	58.3	8.6	22.5
0.190	N	50.5	16.2	10.2	60.7	26.4	64.0	54.0	3.3	27.6
0.230	N	45.0	21.9	10.2	55.2	32.1	62.4	54.4	7.2	22.3
0.365	N	32.3	15.6	10.2	42.5	25.8	58.6	49.4	16.1	23.6
0.450	N	27.4	9.7	10.2	37.6	19.9	56.9	47.1	19.3	27.2
0.585	N	23.1	2.4	10.2	33.3	12.6	56.0	46.0	22.7	33.4
0.715	N	20.3	11.4	10.2	30.5	21.6	56.0	46.0	25.5	24.4
0.805	N	20.7	12.6	10.2	30.9	22.8	56.0	46.0	25.1	23.2

To make insure that the EMI Filter works fine I have tested another sample

S2 test result

- **< Disturbance Voltage of AC Mains port L and N Terminal >**

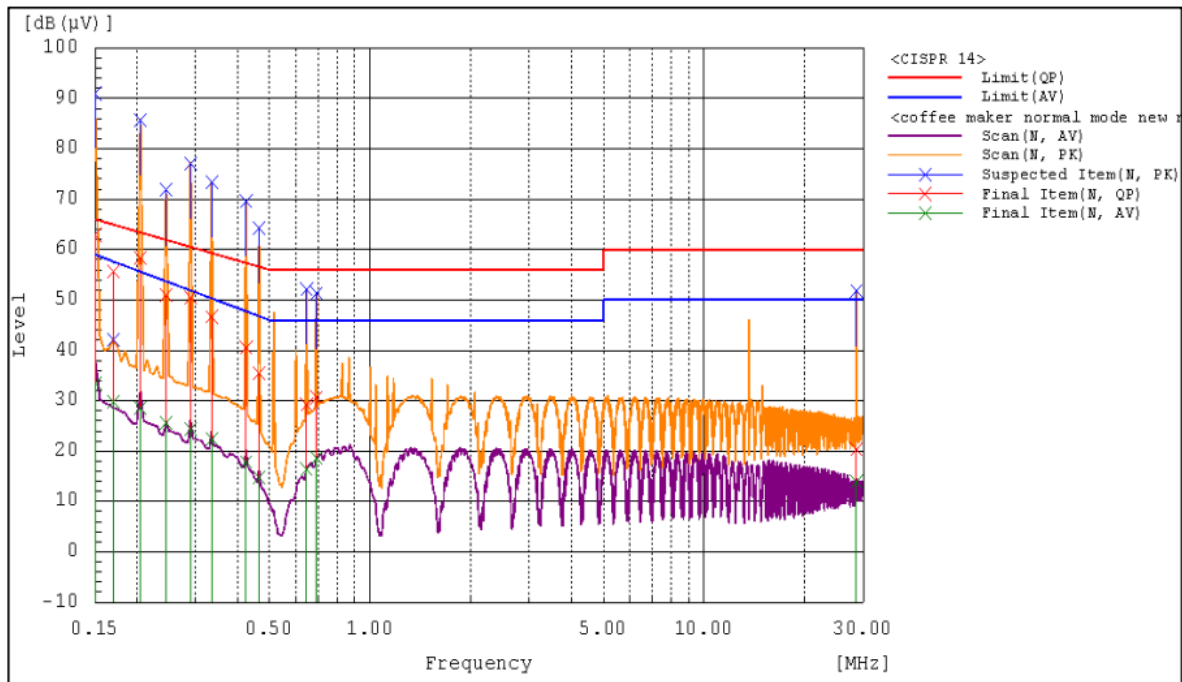
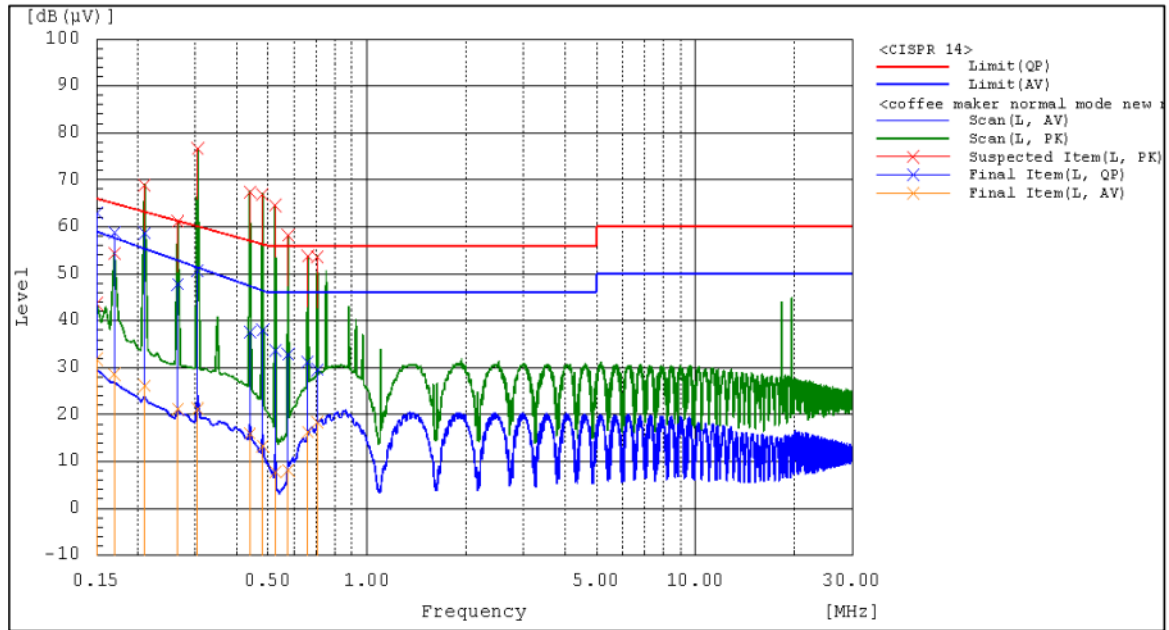


Fig 2: Disturbance Voltage spectrum of N-Terminal.

Final Items Data

Frequency [MHz]	Line	Reading QP [dB(μV)]	Reading AV [dB(μV)]	Factor [dB]	Level QP [dB(μV)]	Level AV [dB(μV)]	Limit QP [dB(μV)]	Limit AV [dB(μV)]	Margin QP [dB]	Margin AV [dB]
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0.165	L	46.7	26.1	10.2	56.9	36.3	65.2	58.0	8.3	21.7
0.190	L	51.0	25.2	10.2	61.2	35.4	64.0	56.4	2.8	21.0
0.235	L	42.4	22.8	10.2	52.6	33.0	62.3	54.2	9.7	21.2
0.280	L	39.6	21.1	10.2	49.8	31.3	60.8	52.3	11.0	21.0
0.365	L	33.2	17.6	10.2	43.4	27.8	58.6	49.4	15.2	21.6
0.500	L	27.5	6.1	10.2	37.7	16.3	56.0	46.0	18.3	29.7
0.545	L	18.9	-4.5	10.2	29.1	5.7	56.0	46.0	26.9	40.3
0.675	L	22.2	11.2	10.2	32.4	21.4	56.0	46.0	23.6	24.6
0.160	N	46.7	25.6	10.2	56.9	35.8	65.5	58.3	8.6	22.5
0.190	N	50.5	16.2	10.2	60.7	26.4	64.0	54.0	3.3	27.6
0.230	N	45.0	21.9	10.2	55.2	32.1	62.4	54.4	7.2	22.3
0.365	N	32.3	15.6	10.2	42.5	25.8	58.6	49.4	16.1	23.6
0.450	N	27.4	9.7	10.2	37.6	19.9	56.9	47.1	19.3	27.2
0.585	N	23.1	2.4	10.2	33.3	12.6	56.0	46.0	22.7	33.4
0.715	N	20.3	11.4	10.2	30.5	21.6	56.0	46.0	25.5	24.4
0.805	N	20.7	12.6	10.2	30.9	22.8	56.0	46.0	25.1	23.2

The result is PASS

Notice: I have raised the X-Cap to 0.33uf in order to enhance the differential mode noise and the result is pass at all frequencies

Reading in the EMC book is highly recommended as it will explain a lot and you will gain high skills

References:



Test report before EMI: coffee maker DV test trial 3 CNC PCB.pdf



Test report after EMI S1: coffee maker DV test trial 10.pdf



Test report after EMI S2: coffee maker DV test trial 10 S2.pdf



EMC Book: Henry Ott - Electromagnetic Compatibility Engineering-Wiley (2009).pdf



EMI Limit: emc limit.pdf



Coil craft EMI Filter design: CM Filter Design.pdf